

Nikolaos Goutzoulis

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EDUCATION

Columbia University
Computer Science MS

Aug 2024 - Dec 2025
New York, NY

- GPA: 3.61/4.00

- Relevant Courses: Unsupervised Learning, Advanced Deep Learning, Advanced Big Data and AI

National Technical University of Athens

Jul 2024

Electrical and Computer Engineering - Integrated MEng

Athens, GR

- Grade: 8.7/10 (top 5% of 400 students)

- Relevant Courses: Deep Learning, Databases, Advanced Algorithms

PROFESSIONAL EXPERIENCE

Ripple

Jan 2026 - Present

Data Scientist

San Francisco, CA

- Part of the Data Core Team.

Columbia University Irving Medical Center

Oct 2025 - Present

Graduate Research Assistant - LLMs

New York, NY

- Building an **agentic AI pipeline with LLM evaluators**, and quality-measurement workflows to detect medical errors from clinician notes. Developing **RAG systems (Pinecone + LangChain)** to support HITL clinical review and producing technical documentation/specifications.

Ripple

May 2025 - Aug 2025

Blockchain Data Scientist Intern

New York, NY

- Built **PySpark/Databricks evaluation pipelines** analyzing 5M+ XRPL DEX transactions, producing standardized benchmarks for AMM slippage, impermanent loss, and LVR used in **production Tableau dashboards**.
- Developed a **Kubernetes-orchestrated multi-agent LLM system** enabling natural-language smart-contract deployment through **backend APIs**, with **task-evaluation logic achieving a 95% success rate** for non-technical users.
- Executed **A/B tests and drift analyses** on real-time price benchmarking pipelines and automated reconciliation workflows in **Python + PySpark**, **reducing manual review effort by 80%** while producing clear documentation for product teams.

EmDoT S.A.

Oct 2023 - Jul 2024

Machine Learning Engineer - Thesis Collaboration

Athens, GR

- Enhanced predictive models with **PyTorch** using **CUDA**, and a random forest with **Sklearn**. Forecasted accident risk and improved prediction accuracy by 15% leading to more **efficient insurance driver allocation** and reducing **insurance company's** operational costs.
- Optimized model performance by 20% through feature engineering and **CUDA optimization, hyperparameter tuning**, improving training efficiency and accuracy. Used **SHAP** to enhance model **explainability**.
- **Engaged directly with clients**, worked with cross-functional teams and **deployed ML models** in industrial use.

Data Science Intern

Jul 2023 - Sep 2023

- Created anomaly detection algorithms for Bin Fill Level data, using **SQL, Tableau, R**, and **predictive statistical models like Prophet and SARIMAX** to model the cost of garbage truck deployment for the municipality. Reduced noise in data by 20% while maintaining 95% of original signal.

PROJECTS

NutritionAI: LLM Nutritional Personalized Assistant

Jan 2025 - Mar 2025

Columbia University

- Deployed **data infrastructure on AWS** for NutritionAI, a personalized LLM nutritional assistant. Built data pipelines for model **fine-tuning (Llama 3 8B with LoRA)**, **used dataset augmentation**, and developed microservices architecture (FastAPI + GraphQL) with CI/CD automation for production deployment.

NutritionAI: LLM Nutritional Personalized Assistant

Jan 2025 - Mar 2025

Columbia University

- Preprocessed a conversational and nutritional dataset with **Python scripts** and fine tuned **Llama3 8B** on AWS's NVIDIA T4 GPU. The natural language processing project produced personalized nutritional programs for each user.

SKILLS

Programming Languages: Python, C++, TypeScript, Matlab, SQL, R

ML Frameworks: Tensorflow, CUDA, PyTorch, Keras, Scikit-learn, Transformers, XGBoost

Cloud, Development and Other Tools: GCP, Databricks, Docker, Kubernetes, AWS (S3, Lambda, SageMaker, EKS), PySpark, API design, Matplotlib, GitHub, Tableau, FastAPI, LoRA/QLoRA, MLFlow (familiar), LangGraph, CI/CD, On-Chain Analytics, Solidity (familiar)